ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Electronics and Digital Concepts for Robotics paper 1 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1 1.1 D ✓ 1.2 B ✓ 1.3 A ✓ 1.4 C ✓ 1.5 B ✓ 1.6 A ✓ 1.7 D ✓ 1.8 C ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 True ✓ 2.4 True ✓ 2.5 False ✓ 2.6 True ✓ 2.7 True ✓ 2.8 False ✓ 2.9 True ✓ 2.10 False ✓ (10)

[10]

QUESTION 3

3 3.1 F ✓ 3.2 D ✓ 3.3 I ✓ 3.4 B ✓ 3.5 J ✓ 3.6 G ✓ 3.7 C ✓ 3.8 A ✓ 3.9 H ✓ 3.10 E ✓ (10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 $4{,}5 \times 10^6$ ✓ (1)

4.2 $0{,}0072$ ✓ (1)

4.3 $a = (v - u) \div t$ ✓ $= (2 - 0) \div 8$ ✓✓ $= 0{,}25\ m/s^2$ ✓ (4)

4.4 $F = m \times a = 5 \times 2$ ✓ $= 10\ N$ ✓ (2)

4.5 $W = F \times s$ ✓ $= 50 \times 3$ ✓ $= 150\ J$ ✓ (3)

4.6 $P = W \div t$ ✓ $= 150 \div 5$ ✓ $= 30\ W$ ✓ (3)

4.7 Proton ✓ – positive ✓; neutron ✓ – no charge (neutral) ✓; electron ✓ – negative ✓ (6)

[20]

QUESTION 5

5.1 5.1.1 Polarised (electrolytic) capacitor ✓ 5.1.2 Potentiometer / variable resistor ✓ 5.1.3 Battery (of cells) ✓ 5.1.4 Ammeter ✓ 5.1.5 Earth (ground) ✓ (5)

5.2 Base line ✓, collector ✓ and emitter ✓ lines; arrow on the emitter pointing INWARDS (towards the base) ✓; terminals labelled B, C, E ✓ (5)

[10]

QUESTION 6

6.1 A conductor has many free electrons and lets current flow easily ✓ – e.g. copper, aluminium, silver ✓. An insulator has very few free electrons and does not let current flow ✓ – e.g. rubber, PVC, glass, wood ✓. (4)

6.2 $22 \times 1\,000 = 22\ \text{k}\Omega\ (22\,000\ \Omega)$ ✓✓; tolerance $\pm$ 10% ✓ (3)

6.3 $\text{R}_\text{T} = (\text{R}_1 \times \text{R}_2) \div (\text{R}_1 + \text{R}_2)$ ✓ $= (100 \times 300) \div 400$ ✓ $= 75\ \Omega$ ✓ (3)

[10]

QUESTION 7

7.1 $16 + 0 + 4 + 2 + 0$ ✓ $= 22$ ✓ (2)

7.2.1 Correct OR symbol ✓; truth table ✓✓✓✓ (00 → 0, 01 → 1, 10 → 1, 11 → 1) (5)

7.2.2 Correct EX-OR symbol (OR shape with an extra curved line at the inputs) ✓; truth table ✓✓✓✓ (00 → 0, 01 → 1, 10 → 1, 11 → 0) (5)

7.3 An analogue signal changes smoothly and can have any value in a range ✓ (e.g. a sine wave from a microphone) ✓; a digital signal has only two levels, 0 and 1 (e.g. a square wave) ✓. (3)

[15]

QUESTION 8

8.1 Any THREE types ✓✓✓: water; foam; carbon dioxide (CO_2); dry chemical powder (DCP). Electrical fires: CO_2 or dry powder ✓ (4)

8.2 Any FIVE in a sensible order ✓✓✓✓✓: do not touch the person while they are still in contact with the supply; switch off the power at the switch / breaker; if that is not possible, move the person away with a dry non-conducting object (wooden broom); call for help / emergency services; check breathing and pulse and start CPR if needed; treat burns with cool water and cover them; keep the person warm and still until help arrives. (5)

8.3 Any THREE tools ✓✓✓ with their uses ✓✓✓, e.g. soldering iron – joins components with solder; side cutters – cut wires and component leads; wire strippers – remove insulation; long-nose pliers – hold and bend small parts; multimeter – measures voltage, current and resistance; desoldering pump – removes solder. (6)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE